Break Into Aerospace

From Electrical Engineering to Aerospace

Career Transition Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: Electrical engineers are among the most consistently recruited non-aerospace graduates into the aerospace industry — and among the most consistently underprepared for their interviews. Not because their skills are wrong for the sector. Precisely the opposite: aerospace has a specific and persistent shortage of engineers who understand high-reliability embedded systems, power electronics, RF systems, digital signal processing, and electric machine design. The problem is that electrical engineers applying to aerospace roles often do not know which of their skills are directly valuable, which require translation, and which gaps they need to fill before the interview. They arrive with strong technical foundations and weak aerospace contextualisation, which interviewers read — correctly — as insufficient preparation for a domain where context determines whether a technically correct answer is also an operationally correct one.

This guide maps the translation layer between electrical engineering and aerospace. It identifies the subspecialties of aerospace where EE backgrounds are strongest, names the specific gaps that need to be filled, and provides a structured preparation pathway for the interview. It is written for electrical engineers at any career stage — from recent graduates with a degree in EE or electronics who want to enter aerospace from the start, to experienced power electronics or embedded systems engineers who are mid-career and want to move into a sector with stronger safety culture and longer-horizon engineering.

The aerospace roles most accessible from an EE background — in roughly descending order of how directly EE skills transfer — are: avionics systems engineering and certification (where digital systems, ARINC bus architectures, DO-178C software, and DO-254 hardware are the daily vocabulary), electric propulsion engineering (Hall thrusters, power processing units, battery systems management), electric and hybrid-electric aviation (eVTOL power electronics, motor drive systems, battery management), spacecraft power systems engineering (solar array regulators, battery charge/discharge management, power distribution), radar and electronic warfare systems engineering (RF circuits, phased array signal processing, waveform design), GNSS receiver engineering (signal processing, correlation, tracking loops), spacecraft attitude determination and control systems (sensor signal processing, filter design, actuator drive electronics), and in-space propulsion power processing. A secondary tier includes aircraft structural health monitoring (sensor networks, signal conditioning), flight test instrumentation (data acquisition, sensor calibration), and CNS/ATM ground systems engineering (RF propagation, ILS and VOR system maintenance).

Part I: What Aerospace Employers Actually See in an EE Candidate

The Genuine Advantages — What Transfers Directly

Aerospace employers know that electrical engineers bring specific technical foundations that aerospace graduates frequently lack. The signal processing fluency — Fourier transforms, Z-transforms, filter design, spectral estimation, sampling theory — that is core EE curriculum is directly applicable to radar signal processing, GNSS receiver algorithm design, and sensor data fusion in avionics. The control theory foundation — state space, root locus, Bode plots, stability margins, discrete-time implementation — that every control systems EE module teaches is exactly the analytical toolkit of flight control system design. The embedded systems competence — real-time operating systems, interrupt league management, memory-mapped I/O, hardware-software co-design — that EE graduates develop through microcontroller projects and digital systems coursework is the foundation of avionics firmware development. The power electronics knowledge — switching converters, motor drives, inverter topology, EMI mitigation — that power electronics EEs bring is directly applicable to electric propulsion systems, eVTOL power electronics, and spacecraft power management.

These advantages are real, and they are not merely theoretical. An EE with a strong control theory background arriving in an aircraft flight control system group is ahead of a mechanical engineering graduate in the tools they need for their first design task. An EE with power electronics experience arriving in an eVTOL inverter design team is more immediately productive than any other engineering background. The translation task is not to become an aerospace engineer from scratch — it is to contextualise existing skills in the aerospace domain and fill the specific gaps that aerospace imposes.

The Specific Gaps — What Does Not Transfer Without Work

The safety methodology gap. Aerospace safety engineering — DO-178C for software, DO-254 for complex electronic hardware, ARP4761/4754A for system safety assessment, and the regulatory framework of CS-25 / FAR Part 25 — is a specific body of method that has no direct equivalent in consumer electronics, automotive (though ISO 26262 is the closest analogue), or telecommunications. The concept of a Design Assurance Level derived from a failure condition severity analysis, the requirement for independent verification of safety-critical software, and the development lifecycle documentation requirements of a Part 25 avionics programme are genuinely new to most EE graduates. This is the most commonly cited reason EE-background candidates fail aerospace interviews for avionics and systems roles: they describe their embedded systems work correctly but cannot explain how it would be done differently in a DO-178C DAL A context.

The aerostructural physics gap. Electrical engineers understand electromagnetic fields. They do not necessarily understand aerodynamic loads, fatigue, damage tolerance, or the relationship between structural stiffness and aeroelastic flutter. For roles in avionics systems integration, this matters: a flight control computer that introduces computational delay must not excite structural modes; an antenna mounted on the fuselage must not compromise the structural integrity of the skin panel it is attached to; a sensor wiring loom must be routed to avoid mechanical damage during wing flex. The EE who does not understand that the aircraft bends, vibrates, and experiences acoustic loads will design electrical systems that are technically correct in isolation and wrong in installation.

The regulatory and certification framework gap. CS-25 / FAR Part 25, DO-178C, DO-254, ARP4761, and the EASA / FAA approval processes for avionics equipment (the ETSO / TSO process) are the regulatory vocabulary of aerospace electronics. An EE who has designed embedded systems for consumer or industrial applications has never operated within this framework. Understanding what a Technical Standard Order (TSO) authorisation means, how avionics equipment obtains EASA or FAA approval, what the relationship between a product's Development Assurance Level and its failure condition classification is, and what the testing and documentation requirements are for a DAL A avionics LRU (Line Replaceable Unit) is the regulatory gap that every EE entering aerospace needs to close.

The operational context gap. Electronics that operates correctly in a laboratory at 25°C and sea level may fail at -55°C at 40,000 feet, in a high-altitude electromagnetic environment, after 20,000 hours of operation with no possibility of maintenance access. The temperature range, the altitude (and associated low-pressure reduced dielectric strength), the vibration spectrum from the airframe, the EMI environment from other aircraft systems, and the required operational life all impose design constraints that consumer and industrial electronics typically do not face. The EE who understands these constraints — who has read DO-160 (Environmental Conditions and Test Procedures for Airborne Equipment) and understands what the temperature, vibration, EMI, and altitude test categories require — has filled the most immediately visible gap.

The systems thinking gap. In many EE roles — circuit design, PCB layout, firmware development — the engineer is responsible for a subsystem whose interfaces with the broader system are relatively well-defined. In aerospace, the system is always the governing context. A flight control computer's timing behaviour affects the aircraft's stability margins. An avionics bus architecture's latency affects sensor fusion accuracy. A battery management system's protection thresholds affect the propulsion system's authority in a degraded mode. EE engineers who have operated primarily in subsystem roles need to explicitly develop the habit of reasoning about how their subsystem's behaviour propagates into the aircraft-level consequence, because aerospace interviewers will probe this directly.

By Career Stage

Recent EE Graduate
The recent EE graduate has the advantage of fresh, strong technical fundamentals and the disadvantage of no work experience to draw on for the specific aerospace application of those fundamentals. The transition gap is primarily contextual — understanding where their signal processing, control theory, and embedded systems knowledge applies in aerospace — rather than technical. The preparation task is targeted: close the DO-178C and DO-254 literacy gap, learn the ARINC bus architectures (ARINC 429, ARINC 664/AFDX) that define avionics data exchange, and build familiarity with at least one specific aerospace domain in enough depth to discuss a real application.

Mid-Career EE (5–15 years)
The mid-career EE has a technical track record that aerospace employers will assess at the level of the real systems they have built — what hardware they have designed, what software they have written, what EMI they have debugged, what production reliability issues they have resolved. The transition task is to translate this track record into aerospace terms: to show that the motor controller they designed for an industrial robot would be designed differently — and documented differently, and tested differently — in an eVTOL context, and to demonstrate that they understand why. The preparation task is heavier on the safety methodology (DO-178C, DO-254, ARP4754A) and the operational environment (DO-160) because these are where the mid-career EE's assumptions about how electronics engineering is done will be most directly challenged.

Senior EE / Technical Lead
The senior EE who has led embedded systems teams, managed development programmes, and interfaced with regulatory bodies (FCC, CE marking, IEC 61508 for functional safety) has a strong foundation for aerospace senior roles — but must specifically demonstrate that they understand the aerospace safety framework and can apply it to their domain. The preparation task includes the safety methodology gap closure (DO-178C at implementation depth, not just conceptual description), the regulatory vocabulary (what an EASA ETSO authorisation requires versus a CE mark), and at least one specific aerospace application domain studied to the depth of a mid-level practitioner.

Domain Mapping — Where Your EE Background Fits Best

Power Electronics → Electric Propulsion / eVTOL / Spacecraft Power
Motor drive design, switching converter topology, SiC MOSFET characterisation, EMI filter design, thermal management of power semiconductors — all transfer directly. The aerospace-specific addition: the weight budget (every gram of heat sink is payload), the DO-254 hardware assurance requirements for the gate driver and protection circuits, and the failure mode and effects analysis that demonstrates no single electronics failure causes a propulsive failure condition of unacceptable severity.

Embedded Systems / RTOS → Avionics Software / Flight Control Computers
Real-time scheduling, interrupt latency, hardware abstraction layers, SPI/I2C/UART peripheral integration — these transfer. The aerospace-specific addition: DO-178C at the level of development assurance (not just "we tested it" but "we verified every requirement through a traceable test, with structural coverage demonstrated at the required level for the assigned DAL"), and the ARINC 429 and ARINC 664/AFDX bus interfaces that are the lingua franca of commercial avionics.

RF / Microwave / Signal Processing → Radar, GNSS, CNS/ATM
Filter design, ADC/DAC interfacing, matched filter implementation, FFT-based spectrum analysis — these transfer to radar signal processing and GNSS receiver design with minimal translation. The aerospace-specific addition: the specific signal structures (GPS L1 C/A, Galileo E1 OS — their PRN codes, their data message formats), the specific regulatory framework (DO-229 for SBAS airborne receivers, TSO-C145/C146 for GNSS receiving equipment), and the specific error modelling (ionospheric delay, multipath, receiver noise) that determines navigation performance.

Control Systems → Flight Control / ADCS / Electric Propulsion Control
State space, discrete-time implementation, Kalman filtering, linear quadratic regulation — these transfer directly to spacecraft attitude control and to electric propulsion throttle control. The aerospace-specific addition: the handling qualities requirements that define what the controlled system must feel like to a pilot

(CS-25.143 and MIL-HDBK-1797), the structural mode notch filtering requirement that prevents the FCS from exciting flexible body modes, and the aeroelastic coupling between the control law and the aircraft's structural dynamics.

Power Systems / Power Management → Spacecraft Power / Battery Systems
Battery management, DC-DC conversion, bus regulation, fault protection and isolation — these transfer to spacecraft power system engineering. The aerospace-specific addition: the space environment (radiation effects on semiconductors, thermal cycling from eclipse to sunlight, atomic oxygen effects on exposed materials), the qualification test sequence (vibration, thermal vacuum, radiation), and the specific reliability requirements that flow from the mission's failure condition classification.

Part II: 20 Specific Prep Tasks for the Transition

FOUNDATIONAL — DO THESE FIRST REGARDLESS OF TARGET ROLE

8–10 hours

1. Understand DO-178C at implementation depth — the DAL system, the objectives, and what independence actually requires
Skill: The software assurance standard that governs all safety-critical avionics software — the single largest gap for EE-background candidates
Why the field cares: DO-178C (Software Considerations in Airborne Systems and Equipment Certification) is the software development assurance standard for civil aviation. Every EE who has written embedded software — firmware for a motor controller, a DSP algorithm for a signal processing chain, a real-time control loop for an industrial application — has written software that would require DO-178C treatment if it were performing a safety-critical function in a certificated aircraft. DO-178C's fundamental innovation is the Design Assurance Level (DAL) system: the required rigour of the software development process is determined by the severity of the failure condition of the function the software implements. DAL A (Catastrophic failure condition) requires the highest rigour — independent verification, structural coverage at the modified condition/decision coverage (MC/DC) level, and formal requirements-based testing with complete traceability. The EE who can explain what MC/DC coverage requires (every condition in every decision has taken both a true and false value, and each condition independently affects the decision outcome), why independence means the verification team cannot be the development team, and what the planning documents (PSAC, SDP, SVP, SCMP) represent is demonstrating DO-178C literacy at a level that most EE-background candidates do not achieve.
Done signal: You can explain what a DAL A avionics software development process requires in terms of planning documents, development standards, verification activities, and structural coverage — and contrast this with a DAL D process. You can explain what MC/DC coverage means and why it is required at DAL A specifically. You can describe how the DAL is derived from the system safety assessment (from the FHA failure condition severity, through the PSSA architecture analysis, to the function DAL assignment). You can describe one aspect of DO-178C that would change how you developed a motor controller firmware — even one you previously developed correctly — in a DAL B avionics context.

5–6 hours

2. Learn DO-254 — the hardware assurance equivalent for complex electronic hardware
Skill: The hardware development assurance standard for programmable logic and custom ICs in avionics
Why the field cares: DO-254 (Design Assurance Guidance for Airborne Electronic Hardware) applies to complex electronic hardware — FPGAs, ASICs, and programmable logic devices — used in certificated aircraft systems. The EE who has designed FPGA-based signal processing systems, DSP hardware, or custom SoC designs for industrial or defence applications is the most valuable target hire for avionics hardware development teams — but only if they understand how their hardware development process must change for DO-254 compliance. DO-254 imposes requirements on hard-

5–6 hours

ware requirements capture, conceptual design, detailed design, implementation, verification (simulation, analysis, and physical test), and configuration management that have no direct equivalent in consumer or industrial FPGA development. The specific requirement for hardware-software separation (the FPGA firmware is subject to DO-254 hardware assurance, not DO-178C software assurance — a distinction that surprises many EEs) and the requirement for independent verification at the higher DALs are the DO-254-specific knowledge items that avionics hardware development interviews probe.
Done signal: You can explain the scope of DO-254 — what types of hardware it applies to (complex electronic hardware: FPGAs, ASICs, PLDs) and what it does not apply to (simple electronic hardware: standard parts, simple programmable devices below the complexity threshold). You can describe the DO-254 development lifecycle phases and what each phase produces. You can explain why FPGA firmware is subject to DO-254 rather than DO-178C in most avionics contexts, and describe what independent verification means for a DAL B FPGA design. You can identify one specific element of your existing FPGA development practice that would need to change for DO-254 compliance.

5–6 hours

3. Understand DO-160 — the environmental test standard for airborne equipment
Skill: The operating environment specification that determines what your electronics must survive
Why the field cares: DO-160 (Environmental Conditions and Test Procedures for Airborne Equipment) defines the temperature, altitude, humidity, vibration, shock, acoustic noise, EMI, lightning, icing, and power input conditions that all avionics equipment must be demonstrated to survive. It is the aerospace equivalent of MIL-STD-461 (EMI), IEC 60068 (environmental testing), and automotive AEC-Q100 (automotive qualification) combined into a single standard. The EE designing electronics for aerospace — whether an avionics LRU, an electric propulsion inverter, or a spacecraft power converter — must design to DO-160 from the beginning because the test categories impose design constraints: the altitude category (operating at 40,000 feet means the air pressure is approximately 18 kPa, which reduces the breakdown voltage of air gaps and requires derating of clearance and creepage distances), the vibration category (which determines the mechanical stress on PCB-mounted components and connector contacts), and the EMI category (which determines the filtering and shielding requirements for both conducted and radiated emissions and susceptibility). The EE who has never designed for DO-160 will design circuits that pass a laboratory test and fail a qualification test, and the gap between the two is both expensive and schedule-critical.
Done signal: You can describe the six most safety-relevant DO-160 test categories for avionics electronics — temperature (operating range, rate of change), altitude (maximum operating and storage altitude and their implication for clearance/creepage), vibration (the specific vibration curves relevant to fuselage-mounted versus wing-mounted equipment), EMI (conducted emission and susceptibility limits and what they require from power supply filtering), lightning (the direct and indirect effects categories and what protection is required at each level), and power input (voltage transients, abnormal voltage range, interruptions). For each, you can describe one specific design decision it would impose on an electronics design you have done.

4–5 hours

4. Learn the ARINC bus standards — ARINC 429, ARINC 629, and ARINC 664/AFDX — and how they differ from industrial bus architectures
Skill: The data bus architectures that define how avionics systems communicate in commercial aircraft
Why the field cares: Every commercial aircraft avionics system communicates through ARINC-defined bus architectures that have no direct equivalent in the industrial or consumer electronics world. ARINC 429 — the dominant legacy standard, a 32-bit unidirectional serial bus operating at 12.5 or 100 kbits/s — is used for sensor data transmission and avionic equipment interfacing on virtually every commercial aircraft from the 1970s to today. ARINC 664 / AFDX (Avionics Full Duplex Switched Ethernet) — the switched Ethernet variant used on the A380 and A350 — brings commer-

4–5 hours

cial IT networking infrastructure to avionics with specific determinism and redundancy extensions. Understanding these buses — their electrical characteristics, their data format (the ARINC 429 32-bit word format with label, SSM, data, and parity fields), their timing, and their failure modes — is the avionics data architecture literacy that EEs entering avionics systems engineering roles must have. The EE who arrives knowing only I2C, SPI, CAN, and Ethernet is missing the specific protocols they will use from their first week.

Done signal: You can describe the ARINC 429 bus electrical characteristics (differential bipolar, $\pm$10V), the 32-bit word structure (label in bits 1–8, SSM in bits 30–31, data in bits 9–28, parity in bit 32), how label determines the data parameter being transmitted, and what the Significant Status Matrix (SSM) field communicates about data validity. You can explain why ARINC 429 is unidirectional (one transmitter, up to 20 receivers) and what the consequence is for bus topology. You can describe what AFDX adds over standard Ethernet for avionics use (bounded latency through Virtual Link traffic shaping, redundant paths for fault tolerance, deterministic delivery guarantees).

5–6 hours

5. Understand the system safety assessment process — FHA, PSSA, SSA, and how DAL flows from failure condition severity

Skill: The safety engineering methodology that determines what development rigour your electronics must meet

Why the field cares: Every avionics system and every electric propulsion system is assessed under ARP4761/4754A to determine the failure condition severity of each function — and the resulting severity classification determines the DAL that the software and hardware must meet. Understanding this flow — from the aircraft-level Functional Hazard Assessment that classifies failure conditions as Catastrophic, Hazardous, Major, Minor, or No Safety Effect, through the Preliminary System Safety Assessment that derives the safety requirements for the system architecture, to the final System Safety Assessment that verifies the implemented system meets all safety requirements — is the safety methodology literacy that every EE entering aerospace safety-critical electronics roles must have. The EE who can connect a motor failure in a distributed electric propulsion system to its failure condition classification, and from there to the DAL required for the motor controller firmware, is demonstrating the end-to-end safety methodology understanding that engineering safety-critical avionics and electric propulsion requires.

Done signal: You can explain the ARP4761 assessment hierarchy — FHA, PSSA, SSA — and what each produces. You can explain how the failure condition severity classification (using the CS-25.1309 criteria) determines the DAL for the associated function. You can describe the probability targets for each severity level (Catastrophic: $< 10^{-9}$/flight hour; Hazardous: $< 10^{-7}$; Major: $< 10^{-5}$) and explain why achieving these through fault tree analysis requires specific architectural choices (redundancy for Catastrophic conditions, independence validation through Common Cause Analysis). You can trace through one specific example: a hydraulic system flight control function, its failure condition severity, the resulting DAL for the associated electronics, and what the DAL requires from the development process.

40–60 hours

6. Learn the aerospace domain most aligned with your EE specialisation — at mid-level depth

Skill: The specific aerospace domain knowledge that connects your EE background to the aerospace role you are targeting

Why the field cares: Every EE entering aerospace must engage with at least one specific aerospace domain at sufficient depth to hold a meaningful technical conversation with the engineers who work in it — not just to describe the domain conceptually but to discuss a real technical problem within it. The specific domain depends on the EE specialisation: power electronics engineers should study electric propulsion (Hall thruster PPU design, battery management for eVTOL) or spacecraft power systems (PCDU architecture, battery charge control); signal processing engineers should study GNSS receivers (PVT solver, tracking loop discriminators, SBAS architecture) or radar signal processing (pulse compression, MTI filtering, CFAR detection); control systems engineers should study flight

40–60 hours

control (CS-25.143, stability margins, handling qualities metrics) or spacecraft ADCS (attitude estimation, momentum management); RF engineers should study CNS/ATM ground systems (ILS, VOR, ADS-B) or radar hardware (phased array beamforming, waveform generation). The depth required is not superficial — the candidate who says "I would be interested in GNSS receiver design because I have signal processing experience" without being able to discuss the acquisition correlator, the DLL discriminator, or the ionospheric correction is describing interest, not preparation.
Done signal: You have studied the specific aerospace domain most aligned with your EE specialisation at the depth of the relevant guide in this series — working through at least six of the prep tasks from that guide, building hands-on project experience in the domain, and reaching a level of technical engagement where you could discuss a specific current technical challenge in that domain with someone who works in it professionally.

4–5 hours

7. Build aerospace-specific hardware design competence — clearance/creepage at altitude, derating, and vibration-resistant design practices
Skill: The practical hardware design knowledge that aerospace imposes on top of general EE circuit design
Why the field cares: The EE who has designed PCBs for consumer or industrial products applies design rules appropriate for ground-level operation at controlled temperatures. Aerospace hardware design requires additional constraints that are not taught in general EE programmes and are not in consumer electronics design guidelines. At 40,000 feet, the air pressure is approximately 18 kPa, reducing the breakdown voltage of air gaps to approximately 20% of its sea-level value — and IPC-2221 clearance tables do not account for this. Component derating — operating resistors at 50% of their rated power, capacitors at 70% of their rated voltage, semiconductors at 80% of their rated current — is mandatory in aerospace electronics to achieve the required reliability at the required operating life. Connector and component selection for vibration resistance — the avoidance of leaded through-hole components on highly vibrated PCBs, the use of conformal coating to prevent solder joint fatigue, the selection of connectors with vibration-resistant retention mechanisms — are the mechanical-electrical integration practices that aerospace electronics design requires.
Done signal: You can describe how IPC-2221 clearance and creepage requirements must be modified for high-altitude operation, specifically what the IPC-2221 altitude derating factor is and how it affects the minimum spacing between high-voltage nodes on an avionics PCB. You can describe the standard derating guidelines applied in aerospace electronics (MIL-HDBK-217 basis for reliability calculation, 50% resistor power derating, 70% capacitor voltage derating). You can identify three specific PCB design practices that improve vibration resistance and explain the mechanical mechanism each addresses.

4–5 hours

8. Understand the avionics equipment approval process — ETSO/TSO, the DO-178C/DO-254 compliance submission, and the differences between EASA and FAA approval
Skill: The regulatory pathway that takes an electronics design from development to certificated avionics product
Why the field cares: An avionics LRU does not enter service on a certificated aircraft simply by being technically correct — it must receive regulatory approval demonstrating that its development process met the applicable assurance standards. The European Technical Standard Order (ETSO) and American Technical Standard Order (TSO) approval processes are the regulatory pathways. Understanding what an ETSO authorisation is (an EASA approval of an article that has been shown to meet minimum performance standards — not aircraft-specific approval but article-level approval), what the compliance documentation package contains (the Accomplishment Summary demonstrating DO-178C and DO-254 objectives have been met, the Software Configuration Index, the Hardware Configuration Index, the test reports), and how EASA and FAA approvals differ (TSO and ETSO are separate national approvals; bilateral agreements between the US and Europe allow most equipment to be dual-approved through a coordinated process) are the regulatory process knowledge that EEs

4–5 hours

in avionics product development roles must have.
Done signal: You can explain what an ETSO/TSO authorisation is and what it means for an avionics article's eligibility for installation on certificated aircraft. You can describe the compliance documentation package that an ETSO applicant submits to demonstrate DO-178C and DO-254 objectives have been met. You can explain the relationship between an ETSO authorisation (article-level approval) and a Supplemental Type Certificate or Form 337 (installation approval on a specific aircraft), and why both are required before an avionics LRU can legally be installed and used on a Part 25 aircraft.

3–4 hours

9. Prepare behavioural answers that translate EE experience into aerospace-relevant stories — connecting your systems to their safety consequences
Skill: The narrative translation that makes EE experience legible to aerospace interviewers
Why the field cares: Aerospace interviewers are not necessarily familiar with the specific products and markets an EE has worked in. The EE who says "I designed the motor controller for our warehouse robot" has given information that means nothing to a flight control hardware engineer. The EE who says "I designed the motor controller for a warehouse robot — it was a three-phase PMSM drive using SiC MOSFETs at 48V bus with a custom gate driver ASIC, operating at 20 kHz switching frequency, designed to a 50,000-hour MTBF requirement with functional safety consideration under IEC 61508 SIL 2 for the emergency stop function — if I were designing an equivalent system for an eVTOL motor, the key differences would be the DO-254 hardware assurance requirement for the gate driver logic, the DO-160 vibration and thermal qualification, and the failure mode analysis that connects motor failure to the aircraft's propulsion failure condition severity" has translated their experience into aerospace terms in a way that aerospace interviewers recognise and value.
Done signal: For each of four specific technical projects in your EE career, you have prepared a translation narrative that: describes the project in EE technical terms, explicitly maps the key technical elements to their aerospace equivalents (the switching converter becomes the PPU; the IEC 61508 SIL 2 analysis becomes the DO-254 DAL B hardware assurance; the CE marking EMI compliance becomes DO-160 Category R), and identifies specifically how the development would differ in an aerospace context. The narrative demonstrates that you understand not just the technical equivalences but the regulatory and safety requirement differences.

20–40 hours

10. Build a portfolio project in your target aerospace domain — a hands-on technical demonstration of aerospace-relevant work
Skill: The most direct signal of genuine aerospace engagement that EE-background candidates can provide
Why the field cares: Aerospace employers hiring EE-background candidates face a specific uncertainty: is this person's interest in aerospace genuine and technically grounded, or is it aspirational and superficial? The most direct way to resolve this uncertainty is a portfolio project — a piece of work that demonstrates technical engagement with a specific aerospace problem at the level of someone who has done real engineering, not someone who has read about it. Accessible portfolio projects for different EE backgrounds include: implementing a simplified GNSS PVT solver in Python from RINEX data (for signal processing EEs); designing and building a three-phase motor controller with SiC MOSFETs and testing its DO-160 equivalent EMI performance (for power electronics EEs); implementing a Kalman filter for sensor fusion of an IMU and GPS receiver on a drone or model aircraft platform (for control systems EEs); or implementing a simplified radar target detection algorithm on an SDR platform using real recorded data (for RF EEs). These projects are achievable in 20–40 hours of focused work, and they produce something specific to discuss in the interview.
Done signal: You have completed a portfolio project in your target aerospace domain that produces a specific technical output — a working code implementation, a tested hardware design, or a documented analysis with results compared to reference data. The project is documented (on GitHub, in a technical report, or in a well-structured PDF) at a level that allows a technical interviewer to review it before the interview. You can spend 10 minutes walking through the project — its objectives,

20–40 hours

your approach, the key technical decisions, the results, and one thing you would do differently in a flight-qualified version.

ROLE-SPECIFIC — TARGET AFTER THE FOUNDATIONAL TASKS

4–5 hours

11. Understand the aerospace-specific aspects of electric machine design — motor topology selection for aerospace, thermal limits, and demagnetisation
Skill: The electric machine knowledge specific to aerospace applications for EEs with motor drive experience
Why the field cares: EEs with motor drive experience often understand motor control (torque control, field weakening, MTPA control) at depth but have less detailed knowledge of the motor design itself — the winding configuration, the magnetic circuit design, and the thermal limits that determine the motor's operating envelope. In aerospace applications (eVTOL, electric propulsion for spacecraft, electric actuators for flight control), the motor design is as much an aerospace engineering problem as an electrical engineering problem: the specific power (W/kg) must be maximised within thermal and electromagnetic constraints that are set by the operating environment, the winding insulation class must be compatible with the expected temperature range including DO-160 temperature extremes, and the permanent magnet material must be selected for resistance to demagnetisation at the maximum operating temperature. The EE with power electronics experience who also understands motor design — at the level of why an aerospace permanent magnet motor uses a high pole count (to reduce the switching frequency requirement and thus switching losses in the inverter), why Litz wire is used for the stator winding (to reduce proximity effect losses at high frequency), and what the consequence of thermal demagnetisation of the rotor magnets is for motor controllability — has the most complete aerospace electric propulsion skill set.
Done signal: You can explain why permanent magnet synchronous motors are the standard for aerospace electric propulsion (high specific power, high efficiency over a wide speed range, deterministic torque control). You can describe what the Curie temperature of common rare-earth magnet materials (NdFeB: ∼310°C, SmCo: ∼720°C) means for motor operating temperature limits and why SmCo is preferred for high-temperature aerospace applications despite its higher cost. You can explain what demagnetisation means for a PMSM — the loss of permanent magnet flux due to excessive temperature or opposing field — and what the consequence is for motor controllability and for the safety case.

5–6 hours

12. Understand spacecraft power systems — the PCDU architecture, eclipse management, and the space environment constraints on power electronics
Skill: The spacecraft power engineering context for EEs with power systems or battery management experience
Why the field cares: Spacecraft power systems — the solar array, the Power Conditioning and Distribution Unit (PCDU), and the battery — form a closed power architecture that must operate autonomously for years in a radiation and thermal environment that degrades every component. The PCDU converts the solar array's variable voltage output to the regulated spacecraft bus voltage (typically 28V, 50V, or 100V), manages battery charging and discharging, and distributes power to all spacecraft loads through a combination of regulated and unregulated outputs with fault protection. The eclipse management challenge — maintaining bus voltage using battery energy for up to 40 minutes per orbit in LEO while the solar array is shaded — drives the battery sizing and the charge control algorithm. The radiation environment degrades power electronics semiconductors through total ionising dose and single-event effects, requiring radiation-hardened or radiation-tolerant component selection and mitigation strategies (triple modular redundancy for SEU-sensitive logic, ECC memory for radiation-affected registers). The EE with battery management and DC-DC converter experience is highly relevant to spacecraft power systems — but must understand these space-specific additions.

5–6 hours

Done signal: You can describe the PCDU architecture — the solar array shunt regulator, the main bus regulator, the battery charge/discharge controller, the power distribution switches, and the telemetry monitoring — and explain the function of each element. You can explain the eclipse energy balance calculation — how the battery capacity is sized to provide the required load power during the maximum eclipse duration with a specified depth of discharge margin. You can describe what total ionising dose means for a MOSFET in a spacecraft power converter, and what design approaches (gate oxide hardening, thicker gate oxide, radiation-hardened foundry process) mitigate TID degradation.

4–5 hours

13. For avionics roles: understand the ARINC 653 RTOS standard and partitioned software architecture

Skill: The real-time operating system standard for integrated modular avionics

Why the field cares: Modern commercial aircraft avionics uses Integrated Modular Avionics (IMA) — a single computing platform hosting multiple avionics applications partitioned from each other by the ARINC 653 real-time operating system standard. ARINC 653 provides spatial partitioning (each application has its own protected memory space and cannot corrupt another application's data) and temporal partitioning (each application is allocated a fixed time slot in the schedule and cannot exceed it, preventing one application from starving another of CPU time). This architectural approach — used on the A380, A350, B787, A220, and all modern commercial aircraft — is very different from the single-application RTOS or bare-metal embedded systems that most EE embedded developers have worked with. Understanding what ARINC 653 provides and what it requires from the avionics application developer (the application must be designed to complete its required functions within its allocated time budget; it cannot dynamically allocate memory; its communication with other partitions is through well-defined port interfaces) is the IMA architecture literacy that avionics software developers must have.

Done signal: You can explain what ARINC 653's spatial partitioning provides — the memory protection mechanism that prevents one application from reading or writing another application's memory — and why this is required for multi-DAL-level applications hosted on a single IMA platform. You can explain what temporal partitioning provides — the guaranteed time slot allocation — and what the consequence is for application design (the application must complete all required processing within its allocated time budget; excess budget goes unused rather than being reallocated). You can describe the ARINC 653 port mechanism — sampling ports and queuing ports — and explain how they implement the inter-partition communication that replaces the direct function calls and shared memory that single-partition embedded developers use.

8–12 hours

14. For GNSS/navigation roles: implement a simplified signal acquisition and tracking chain

Skill: The hands-on signal processing project that demonstrates GNSS receiver engineering engagement

Why the field cares: GNSS receiver design is fundamentally a signal processing engineering discipline — the acquisition correlator, the code tracking DLL, the carrier tracking PLL, and the navigation data decoder are all signal processing algorithms implemented in either FPGA hardware or embedded software. The EE with digital signal processing experience — who has implemented FIR filters, FFT-based spectrum analysis, and correlation algorithms — has the specific computational tools that GNSS receiver design requires. The gap is in the specific signal structures (the GPS L1 C/A PRN code, the navigation message format) and the specific algorithm choices (the discriminator function that converts tracking error to a phase correction, the loop filter bandwidth that balances noise and dynamic response). Implementing a simplified acquisition and tracking chain — starting from publicly available GNSS software receiver frameworks (GNSS-SDR, SoftGNSS) and working through the algorithm design to understand what each block does — is the hands-on signal processing project that most directly prepares an EE for a GNSS receiver engineering interview.

Done signal: You have implemented, modified, or deeply studied a GNSS acquisition and tracking chain — either using an open-source GNSS software receiver framework or building simplified

8–12 hours

versions of key blocks from scratch. You can explain what a BPSK spread spectrum signal is and why GNSS signals are received below the thermal noise floor (the link budget gives approximately -130 dBm at the receiver, approximately 20 dB below the thermal noise floor for a 1 MHz receiver). You can explain the correlation gain that the acquisition process recovers (the spreading gain equals 10log(N) where N is the number of chips in the PRN code), what the acquisition uncertainty space is (code phase uncertainty × Doppler frequency uncertainty), and how the acquisition threshold is set from the required probability of detection and false alarm.

5–6 hours

15. For electric propulsion roles: build a detailed understanding of Hall thruster power electronics and the PPU design challenge

Skill: The power electronics interface between the spacecraft and the Hall thruster — the specific domain where power electronics EEs add most value

Why the field cares: The Power Processing Unit for a Hall thruster is the most demanding power electronics design challenge in commercial spacecraft: it must supply a 200–400V discharge supply at 0.5–5A to a load with negative impedance characteristics (the thruster's discharge voltage decreases as discharge current increases, producing the equivalent of a negative resistance that destabilises a standard voltage-controlled power supply), a precise magnetic coil current supply at low ripple, a cathode heater and keeper supply, and a propellant flow controller — all while achieving >90% efficiency in a volume and mass budget of perhaps 2 kg. The negative impedance discharge supply is the specific design challenge that most distinguishes PPU design from other power electronics: a standard peak-current-mode buck converter becomes unstable when connected to a negative impedance load, and the control architecture must include specific loop compensation that accounts for the thruster's dynamic impedance. This is the specific PPU engineering challenge that every power electronics EE entering electric propulsion should understand before their first interview.

Done signal: You can explain what the Hall thruster's negative discharge impedance is — why the discharge current increases when discharge voltage decreases, creating a dynamic negative impedance — and why this characteristic destabilises a standard voltage-mode-controlled boost converter. You can describe the control architecture modification required — current-mode control with appropriate compensation to provide sufficient phase margin when connected to the negative impedance load — and explain why the stability analysis must include the thruster's dynamic impedance model, not just the converter's open-loop characteristics. You can estimate the required PPU efficiency for a 1 kW Hall thruster system on a spacecraft with a 6 kW solar array, and explain what the consequence of a 5% PPU efficiency reduction is for the spacecraft's power budget.

4–5 hours

16. For flight control and ADCS roles: understand the structural coupling challenge — how your control loop can destabilise the aircraft

Skill: The aeroelastic interaction that distinguishes flight control design from any other control system application

Why the field cares: Flight control systems and spacecraft attitude control systems operate on vehicles that are not rigid bodies — they flex, vibrate, and have structural resonances at frequencies that may be within or close to the control system's bandwidth. A flight control law that achieves excellent rigid-body handling qualities but has insufficient attenuation at the first wing bending frequency can couple energy into the structural mode, driving a structural oscillation that the aircraft cannot tolerate. The structural mode notch filter — placed in the feedback path at the structural resonance frequency to attenuate the signal before it re-enters the control loop — is the standard solution, but its design introduces a phase lag at frequencies below the notch that erodes the rigid-body stability margins. The trade between structural coupling attenuation and rigid-body control quality is the aeroservoelastic design challenge that every control systems EE entering flight control engineering must understand. The equivalent in spacecraft ADCS is the flexible appendage coupling problem — a large solar array or antenna boom has a resonance that the attitude control system must not excite — which is a structurally equivalent problem with a spacecraft-specific set of constraints.

4–5 hours

Done signal: You can explain what servo-aeroelasticity is — the feedback loop between control surface actuation, structural deformation, sensor measurement, and further control surface actuation — and describe what happens physically when this loop becomes unstable. You can explain how a notch filter in the control law signal path attenuates structural resonance frequencies, and why it introduces phase lag at frequencies below the notch that affects the rigid-body stability margins. You can describe the trade between notch depth and notch bandwidth — deeper attenuation of the structural mode requires a narrower notch, which introduces less phase lag but is more sensitive to structural mode frequency uncertainty. You can describe the equivalent problem in spacecraft ADCS with a flexible appendage.

2–3 hours

17. For radar and EW roles: understand the specific regulatory and classification environment
Skill: The security and export control landscape that governs aerospace RF and EW engineering
Why the field cares: Radar and electronic warfare roles at aerospace primes and defence contractors are among the most classification-sensitive in the engineering world — and among the most directly accessible from RF/microwave EE backgrounds. The specific challenge for EE candidates: most of the technically demanding radar and EW work is classified, and access to classified programmes requires a security clearance (SC or DV in the UK, Secret or TS/SCI in the US) that takes months to obtain and requires specific nationality and residency criteria. Additionally, radar and EW technology is subject to ITAR (International Traffic in Arms Regulations) in the US and export control regulations in the UK and Europe, which impose restrictions on who can access which technical discussions regardless of clearance. Understanding these constraints before applying — specifically knowing whether you meet the nationality requirements for a clearance, what the timeline is, and which roles at the target employer are clearance-gated versus unclassified — prevents wasted effort and recruiter friction.
Done signal: You understand your own clearance eligibility in your target country (US citizenship for most TS-gated work; UK/dual nationality assessment for UK DV; equivalent in other countries). You know which roles at your target employer are clearance-gated and which are unclassified or require only a lower level of vetting. You have read the relevant export control regulation (ITAR for US employers; UK Export Control for UK defence contractors; EU Dual Use Regulation) to understand what restrictions apply to the specific technology area you are targeting. You can explain to a recruiter what your clearance eligibility is without requiring them to ask.

POSITIONING AND PRESENTATION

3–4 hours

18. Reframe your CV and LinkedIn for aerospace — the translation exercise for every EE role description
Skill: The professional positioning step that determines whether the application gets past the initial screen
Why the field cares: An EE CV written for a consumer electronics or automotive audience will not optimally position the candidate for aerospace — not because the experience is wrong, but because the vocabulary, the emphasis, and the framing are different. An aerospace hiring manager reading about “ISO 26262 ASIL D functional safety analysis” will recognise it as the automotive equivalent of DO-178C DAL A and will appreciate the translation — but only if the candidate makes the comparison explicitly. Similarly, “Designed 3-phase PMSM drive for EV powertrain” does not connect itself to “relevant to eVTOL motor drive design” — the connection must be made by the candidate, explicitly. The CV reframe does not require misrepresentation; it requires translation — using the aerospace vocabulary alongside the domain-specific vocabulary to make the connection legible to an aerospace reader who may not be fluent in the source domain.
Done signal: Your CV and LinkedIn profile have been reframed for your target aerospace role — every significant technical experience has been described using both its original domain vocabulary and the aerospace equivalent, explicitly noting the transferable elements. Your CV summary ex-

3–4 hours

plicitly names the aerospace role you are targeting and connects your EE specialisation to it in one sentence. The skills section explicitly lists DO-178C/DO-254 familiarity (if achieved through the prep tasks above), ARINC bus knowledge, and any domain-specific aerospace knowledge built through the portfolio project.

2–3 hours

19. Prepare a specific answer to "Why aerospace — and what specifically from your EE background makes you ready for this role?"
Skill: The transition motivation and positioning answer that every aerospace EE interview opens with
Why the field cares: Aerospace interviewers hiring EE-background candidates have a specific concern: will this person stay, or will they return to the faster-moving, often higher-paid consumer electronics or automotive sector when the novelty of aerospace wears off? And does their EE background actually prepare them for this specific role, or are they hoping that a general technical competence will substitute for domain knowledge? The answer to "why aerospace?" must address the first concern — with a specific, technical reason why the long development timescales, the rigorous safety methodology, and the domain physics of aerospace are genuinely more interesting to this particular engineer than the alternatives. And the answer to "what from your EE background makes you ready?" must be specific — naming the exact technical elements that transfer directly, acknowledging the gaps, and demonstrating that the preparation work has been done.
Done signal: You have a 2-minute answer that: explains why aerospace specifically — not in terms of childhood passion but in terms of what the engineering environment offers that your current domain does not (a specific example: the DO-178C safety methodology is intellectually more demanding than IEC 61508, and the consequence of the work being done correctly is traceable to the physical safety of specific people in a way that an industrial control system rarely is); names the three most directly transferable elements of your EE background and explains precisely why they apply to the target role; acknowledges the specific gap you have had to close (DO-178C, DO-160, ARINC buses, or whatever is most relevant); and demonstrates that the gap has been addressed (through the portfolio project, the study tasks above, or a specific aerospace engagement). The answer cannot be given unchanged to someone applying for a non-aerospace EE role.

4–6 hours

20. Map your EE specialisation to the specific aerospace employers and roles where it is most competitive — and build a targeted application strategy
Skill: The job market intelligence that determines where to focus preparation effort
Why the field cares: Not all aerospace employers value EE backgrounds equally. The employers most actively recruiting EE specialists from adjacent industries are those working on electric propulsion, avionics hardware development, eVTOL power systems, and GNSS receiver design — domains where the technical challenge is fundamentally electrical engineering applied to an aerospace context. The employers least accessible from a pure EE background without significant aerospace domain knowledge are those working on structural analysis, gas turbine design, aerodynamics, and flight test — where the domain physics is less directly connected to EE fundamentals. Understanding which employers are in the first category, which roles at those employers are most accessible from your specific EE specialisation, and what the interview process at those employers looks like is the job market intelligence that makes a transition job search efficient rather than diffuse.
Done signal: You have identified at least five specific employers where your EE specialisation is directly relevant to open or recurring roles, researched the specific roles at each employer that align with your background, identified the specific technical topics most likely to be assessed at each (based on their job descriptions, LinkedIn profiles of their engineers, and publicly available technical content), and prioritised your preparation effort accordingly. You have at least two specific applications in active progress, each with a personalised CV translation and a portfolio project specifically relevant to the target role.

Summary Reference Table

#	Task Description	
1	DO-178C — DAL system, objectives, independence, MC/DC	All / Do first
2	DO-254 — hardware assurance, FPGA scope, DAL B independent verification	All
3	DO-160 — environmental test categories and their design implications	All
4	ARINC bus standards — ARINC 429 format, AFDX determinism	All
5	ARP4761/4754A safety methodology — FHA, PSSA, SSA, DAL derivation	All
6	Target aerospace domain study — at mid-level depth per the relevant series guide	All / Ongoing
7	Aerospace hardware design — altitude derating, component derating, vibration resistance	All
8	ETSO/TSO approval process — compliance package, EASA vs FAA	All
9	STAR story translation — EE experience reframed in aerospace safety terms	All
10	Portfolio project in target domain — working technical output	All / Critical
11	Electric machine design for aerospace — topology, thermal limits, demagnetisation	Power electronics EEs
12	Spacecraft power systems — PCDU, eclipse management, radiation effects	Power systems EEs
13	ARINC 653 RTOS and IMA architecture — partitioning, ports, time budgets	Avionics / embedded EEs
14	GNSS acquisition and tracking chain implementation	Signal processing EEs
15	Hall thruster PPU design and the negative impedance control challenge	EP / power electronics EEs
16	Structural coupling in flight control and ADCS — notch filters, aeroservoelasticity	Control systems EEs
17	Radar/EW clearance and export control landscape	RF / EW EEs
18	CV and LinkedIn reframe — aerospace translation of every EE experience	

#	Task Description
All	
19	"Why aerospace — what specifically transfers?" — 2-minute answer
All	
20	Targeted employer mapping and application strategy
All	

The Structural Reality of the EE-to-Aerospace Transition

The structural reality of the EE-to-aerospace transition: The EE who wants to move into aerospace is not starting from zero — they are starting from a position that aerospace genuinely needs. The sector has a persistent shortage of engineers who understand power electronics, digital signal processing, embedded systems, and RF systems at the level of someone who has built real hardware and debugged real systems. These skills are not peripheral to aerospace engineering; they are central to the most technically demanding and most rapidly growing parts of the sector — electric aviation, electric propulsion, avionics, GNSS, radar.

The preparation task is not to become an aerospace engineer from scratch. It is to do three specific things: close the regulatory and safety methodology gap (DO-178C, DO-254, ARP4761, DO-160 — the frameworks that determine how aerospace electronics is developed and qualified, which are different in important ways from anything in consumer or industrial electronics); develop genuine depth in one specific aerospace domain (not breadth — depth; the ability to discuss a real technical problem in that domain at the level of a practitioner); and translate the existing EE experience into aerospace terms explicitly (not assuming the interviewer will make the connection between ISO 26262 ASIL D and DO-178C DAL A, but making it explicit).

The EE who does these three things consistently and at genuine depth will, in most cases, be a stronger candidate for the roles where EE skills are most directly relevant — eVTOL power electronics, spacecraft power systems, avionics hardware, GNSS receivers — than many aerospace graduates, precisely because they bring a hardware and systems depth from their previous domain that aerospace programmes do not always develop. The aerospace graduate knows the context. The EE knows the circuits. The most valuable engineer in an eVTOL power electronics team knows both.

That combination — the EE who has done the work to understand the aerospace context — is what this guide is designed to produce. The work is specific, it is achievable in four to eight months of deliberate preparation alongside an existing job, and the result is not an EE who is pretending to be an aerospace engineer but an engineer whose actual skills match the actual needs of the sector they are joining.

Break Into Aerospace